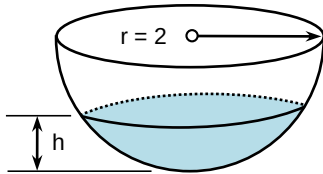


Solutions:

1. **A. 1.0**



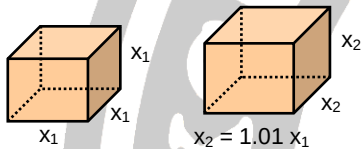
$$V = \frac{\pi h^2}{3} (3r - h)$$

$$5.236 = \frac{\pi h^2}{3} (3(2) - h)$$

$$5 = 6h^2 - h^3$$

$$h = 1 \text{ m}$$

2. **C. 3.03 %**



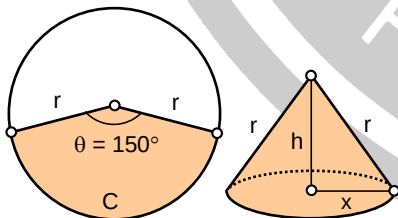
$$\frac{V_2}{V_1} = \left(\frac{x_2}{x_1} \right)^3 = \left(\frac{1.01 x_1}{x_1} \right)^3$$

$$\frac{V_2}{V_1} = 1.0303$$

$$V_2 = 1.0303 V_1$$

Thus, V_2 is increased by 3.03%

3. **C. 13,503.44 cm³**



Let:

C_1 = circumference of the circle

C_2 = circumference of the base of the cone

C = length of arc of the sector

$$C = C_1 - C_2$$

$$r\theta = 2\pi r - 2\pi x$$

$$36 \left(150^\circ \times \frac{\pi}{180^\circ} \right) = 2\pi(36) - 2\pi(x)$$

$$x = 21 \text{ cm}$$

$$h = \sqrt{r^2 - x^2} = \sqrt{36^2 - 21^2} = 29.24 \text{ cm}$$

$$V = \frac{1}{3} \pi x^2 h = \frac{1}{3} \pi (21)^2 (29.24)$$

$$V = 13,503.44 \text{ cm}^3$$

4. **C. 18.85 cm**

$$C = r\theta$$

$$C = 45 \left(24^\circ \times \frac{2\pi \text{ rad}}{360^\circ} \right)$$

$$C = 18.85 \text{ cm}$$

5. **A. 67.5°**

Let:

θ = the angle

$90^\circ - \theta$ = the complement of angle θ

$180^\circ - \theta$ = the supplement of angle θ

$$180 - \theta = 5(90 - \theta)$$

$$180 - \theta = 450 - 5\theta$$

$$4\theta = 270$$

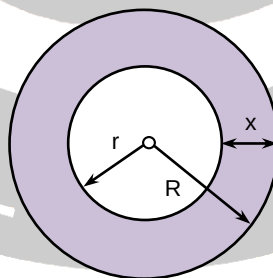
$$\theta = 67.5^\circ$$

6. **B. 170**

$$N = \frac{n}{2}(n-3) = \frac{20}{2}(20-3)$$

$$N = 170 \text{ diagonals}$$

7. **A. 1.76 cm**



Note:

Since half of the area was removed, then the area (A) left is also one-half of the total area.

$$A = \frac{1}{2} [\pi(R)^2] = \frac{1}{2} [\pi(6)^2] = 18\pi$$

But "A" is also equal to the area of the small circle.

(Refer to figure)

$$A = \pi r^2$$

$$18\pi = \pi r^2$$

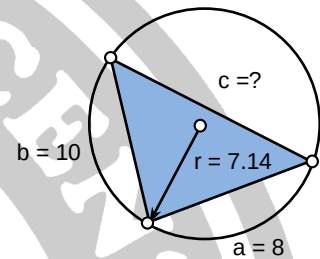
$$r = 4.24 \text{ cm}$$

$$x = R - r$$

$$x = 6 - 4.24$$

$$r = 1.76 \text{ cm}$$

8. **D. 14 cm**

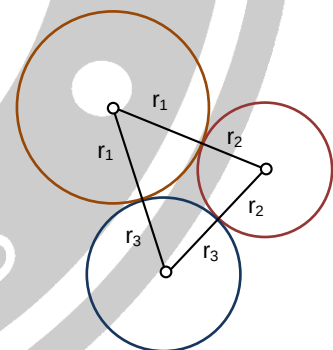


$$A = \frac{abc}{4r}$$

$$39.19 = \frac{8(10)c}{4(7.14)}$$

$$c = 14 \text{ cm}$$

9. **C. 64 π**



$$r_1 + r_2 = 12 \rightarrow \text{Eq.1}$$

$$r_2 + r_3 = 10 \rightarrow \text{Eq.2}$$

$$r_1 + r_3 = 14 \rightarrow \text{Eq.3}$$

Subtract Eq.3 by Eq.2:

$$(r_1 + r_3) - (r_2 + r_3) = 14 - 10$$

$$r_1 - r_2 = 4 \rightarrow \text{Eq.4}$$

Add Eq.4 and Eq.1:



$$(r_1 + r_2) + (r_1 - r_2) = 10 + 4$$

$$2r_1 = 16$$

$$r_1 = 8$$

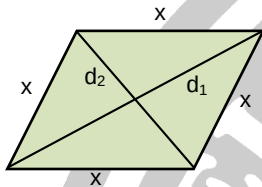
$$r_2 = 12 - 8 = 4$$

$$r_3 = 10 - 4 = 6$$

Let: A = area of the largest circle

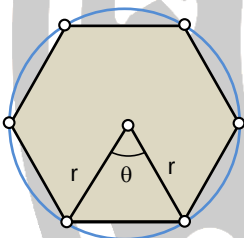
$$A = \pi r_1^2 = \pi (8)^2 = 64\pi$$

10. C. 320 in²



$$A = \frac{1}{2} d_1 d_2 = \frac{1}{2} (32)(20) = 320 \text{ in}^2$$

11. C. 2.598



Let:

A_{total} = area of the hexagon

A_t = area of triangle

$$\theta = \frac{360^\circ}{6} = 60^\circ$$

$$A_t = \left(\frac{1}{2} \right) r^2 \sin \theta$$

$$= \frac{1}{2} (1)^2 \sin 60^\circ$$

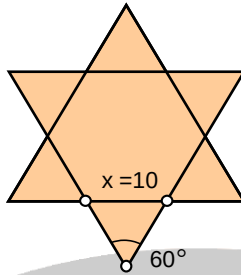
$$A_t = 0.433$$

$$A_{\text{total}} = 6 A_t$$

$$= 6(0.433)$$

$$A_{\text{total}} = 2.598$$

12. C. 519.60 cm²

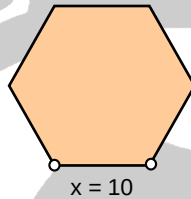


Let:

A₁ = area of the hexagon

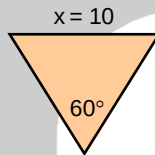
A₂ = area of the triangle

A = total area



$$\theta = \frac{180}{n} = \frac{180}{6} = 30^\circ$$

$$A_1 = \frac{6(10)^2}{4 \tan 30^\circ} = 259.8 \text{ cm}^2$$



By inspection, the triangle must be equilateral.

$$A_2 = \frac{1}{2} x^2 \sin 60^\circ = \frac{1}{2} (10)^2 \sin 60^\circ$$

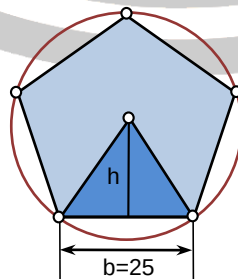
$$A_2 = 43.3$$

$$A = A_1 + 6 A_2$$

$$A = 259.8 + 6(43.3)$$

$$A = 519.6 \text{ cm}^2$$

13. A. 1075 m²



Let:

A_h = area of the hexagon

A_t = area of triangle

$$A_t = \frac{1}{2} bh = \frac{1}{2} (25)(17.2) = 215 \text{ m}^2$$

$$A = 5 A_t = 5(215) = 1075 \text{ m}^2$$

14. B. 24

$$\theta = \frac{(n-2)(180^\circ)}{n}$$

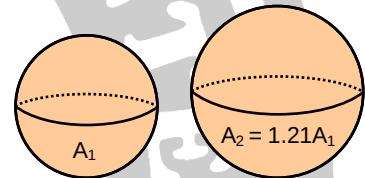
$$165 = \frac{(n-2)(180^\circ)}{n}$$

$$165n = 180n - 360$$

$$15n = 360$$

$$n = 24 \text{ sides}$$

15. D. 33 %



$$\left(\frac{V_2}{V_1} \right)^2 = \left(\frac{A_2}{A_1} \right)^3$$

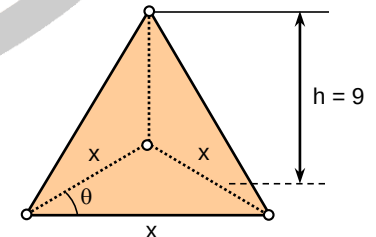
$$\frac{V_2}{V_1} = \left(\frac{A_2}{A_1} \right)^{3/2} = \left(\frac{1.21 A_1}{A_1} \right)^{3/2}$$

$$\frac{V_2}{V_1} = 1.331$$

$$V_2 = 1.331 V_1$$

Thus, V₂ is increased by 33.1%

16. A. 12



Note: $\theta = 60^\circ$, since equilateral triangle.

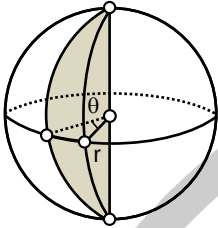
$$V = \frac{1}{3} Bh = \frac{1}{3} \left[\frac{1}{2} x^2 \sin \theta \right] h$$



$$187.06 = \frac{1}{3} \left(\frac{1}{2} \right) x^2 \sin 60^\circ (9)$$

$$x = 12 \text{ m}$$

17. A. 2/3

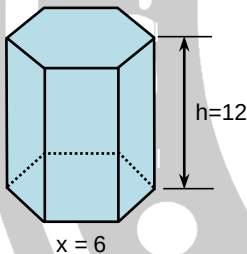


$$V = \frac{\pi R^3 \theta}{270^\circ}$$

$$V = \frac{\pi (1)^3 \left(1 \text{ rad} \times \frac{180^\circ}{\pi} \right)}{270^\circ} = \frac{180}{270}$$

$$V = \frac{2}{3} \text{ cubic units}$$

18. D. 1122.4 cm³



Let:

A = area of one base

x = length of each side of the base

$$A = \frac{nx^2}{4 \tan \theta}$$

$$\theta = \frac{180^\circ}{n} = \frac{180^\circ}{6} = 30^\circ$$

$$A = \frac{(6)(6)^2}{4 \tan 30^\circ}$$

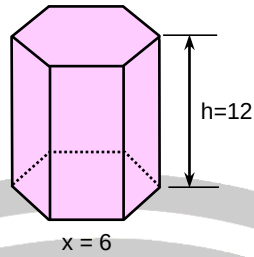
$$A = 93.53 \text{ cm}^2$$

$$V = Ah$$

$$V = 93.53(12)$$

$$V = 1122.4 \text{ cm}^3$$

19. D. 1122.4 cm³



Let:

A = area of one base

x = length of each side of the base

$$A = \frac{nx^2}{4 \tan \theta}$$

$$\theta = \frac{180^\circ}{n} = \frac{180^\circ}{6} = 30^\circ$$

$$A = \frac{(6)(6)^2}{4 \tan 30^\circ}$$

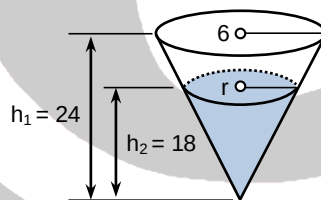
$$A = 93.53 \text{ cm}^2$$

$$V = Ah$$

$$V = 93.53(12)$$

$$V = 1122.4 \text{ cm}^3$$

20. C. 381.70



By ratio & proportion:

$$\frac{6}{24} = \frac{r}{18}$$

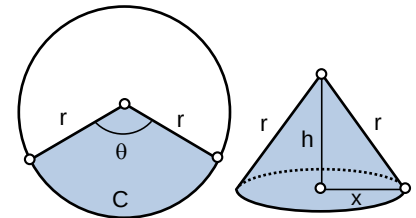
$$r = 4.5$$

$$V = \frac{1}{3} \pi r^2 h_2$$

$$V = \frac{1}{3} \pi (4.5)^2 (18)$$

$$V = 381.7 \text{ cm}^3$$

21. A. 144°



$$x = \sqrt{r^2 - h^2}$$

$$x = \sqrt{(50)^2 - (40)^2}$$

$$x = 30$$

C₁ = circumference of the circle

C₂ = circumference of the base of the cone

C = length of arc

$$C = C_1 - C_2$$

$$C = 2\pi r - 2\pi x$$

$$C = 2\pi(50) - 2\pi(30)$$

$$C = 40\pi$$

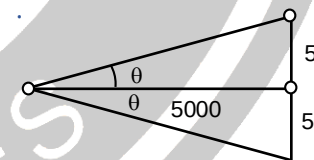
$$C = r\theta$$

$$40\pi = (50)\theta$$

$$\theta = \frac{40\pi}{50} \times \frac{180}{\pi}$$

$$\theta = 144^\circ$$

22. B. 2



$$\tan \theta = \frac{5}{5000}$$

$$\theta = \tan^{-1} \left(\frac{5}{5000} \right)$$

$$\theta = 0.0573^\circ$$

$$2\theta = 2(0.0573)$$

$$2\theta = 0.1146^\circ$$

$$2\theta = 0.1146^\circ \times \frac{6400 \text{ miles}}{360^\circ}$$

$$2\theta = 2.037 \text{ miles}$$



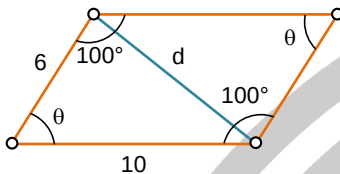
23. D. 995

$$C = r \theta$$

$$C = 300 \left(190^\circ \times \frac{2\pi \text{ rad}}{360^\circ} \right)$$

$$C = 995 \text{ cm}$$

24. C. 10.73



$$\text{Total interior angle} = (n - 2)180$$

$$\text{Total interior angle} = (4 - 2)180$$

$$\text{Total interior angle} = 360^\circ$$

$$2\theta + 100 + 100 = 360$$

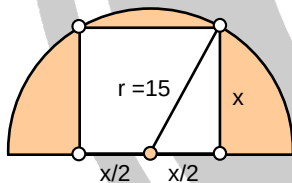
$$\theta = 80^\circ$$

By cosine law:

$$d^2 = 6^2 + 10^2 - 2(6)(10)\cos 80^\circ$$

$$d = 10.73$$

25. C. 173.5 cm²



$$r^2 = x^2 + \left(\frac{x}{2} \right)^2$$

$$15^2 = 1.25x^2$$

$$x = 13.416$$

Let:

A_1 = area of the semi-circle

A_2 = area of the square

A = difference of A_1 and A_2

$$A = A_1 - A_2$$

$$A = \frac{1}{2}\pi r^2 - x^2 = \frac{1}{2}\pi(15)^2 - (13.416)^2$$

$$A = 173.44$$